

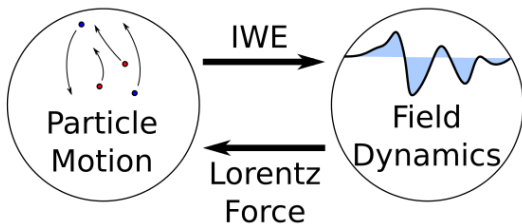
Macroscopic Electrodynamics

Mike Reppert

September 22, 2026

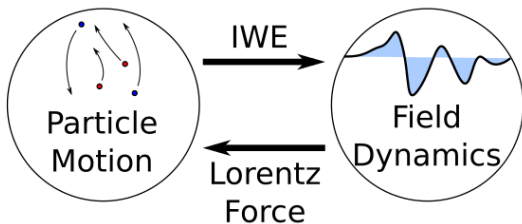
Previously on CHM676...

The Inhomogeneous Wave Equation (IEW – derived from Maxwell's equations) can be used to calculate the EM field *given* particle trajectories.



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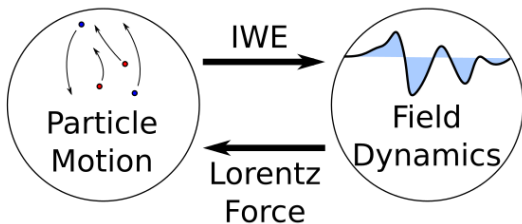
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Today: How to simplify the equations by *coarse-graining*.

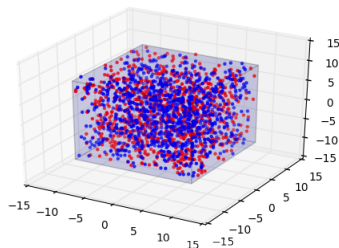
Outline for Today:

- 1 Ensemble Averages and Statistical Mechanics
- 2 Coarse-grained Densities
- 3 Molecular Spectroscopy and Material Polarization

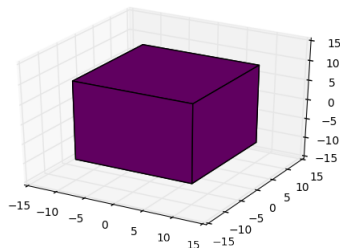
Ensemble Averages and Statistical Mechanics

Statistical Mechanics: Average Physics

Microstate

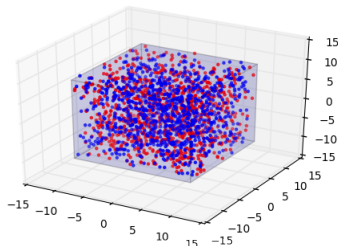


Macrostate

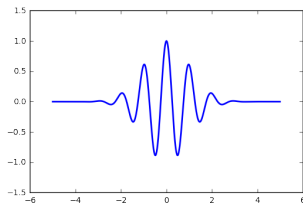
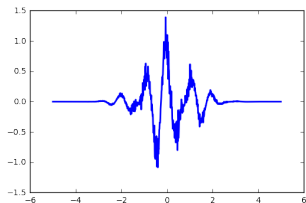
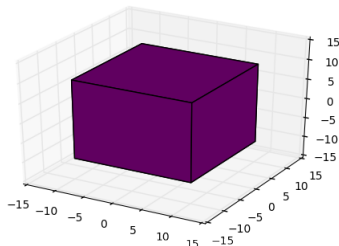


Statistical Mechanics: Average Physics

Microstate



Macrostate



Ensemble Averages

Formally, we define the ensemble average

$$\langle f \rangle_M = \sum_{\mu \in M} p_{\mu}^{(M)} f(\mu)$$

as the average value *over all microstates consistent with a specific macrostate* (temperature, volume, charge, etc.).

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In spectroscopy, we work almost exclusively with the *macroscopic fields*

$$\begin{aligned} \mathbf{E}(\mathbf{x}, t) &\equiv \langle \mathbf{e}(\mathbf{x}, t) \rangle_M \\ \mathbf{B}(\mathbf{x}, t) &\equiv \langle \mathbf{b}(\mathbf{x}, t) \rangle_M . \end{aligned}$$

Macroscopic Fields

What equations determine $\mathbf{E}$ and $\mathbf{B}$?

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Since Maxwell's equations are linear in $\mathbf{b}$ and $\mathbf{e}$:

$$\nabla \cdot \mathbf{E} = 4\pi \langle \rho(\mathbf{x}, t) \rangle_M$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0$$

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c} \langle \mathbf{j}(\mathbf{x}, t) \rangle_M$$

But how do we get rid of the explicit ensemble average?

Wait for it...

Take-Home Points

In statistical mechanics, an *ensemble-averaged* quantity is the average value over all *microstates* consistent with a given *macrostate*.

In spectroscopy, we primarily deal with the *macroscopic fields* $\mathbf{E}$ and $\mathbf{B}$ – the ensemble averages of the *microscopic fields* $\mathbf{e}$ and $\mathbf{b}$.

Since Maxwell's equations are linear, their form is unchanged by ensemble-averaging. The key question is how to handle $\langle \rho \rangle$ and $\langle \mathbf{j} \rangle$.

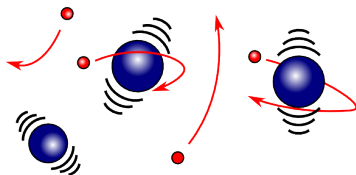
Coarse-grained Densities

Coarse-graining the Densities

A two-step process:

(1) Distinguish between *free* and *bound* charges:

“bound” $\approx$ “stuck to a nucleus”

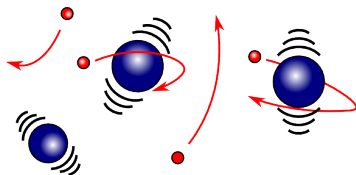


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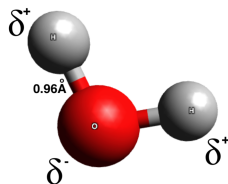
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(2) Multipole expansion in molecular size

- ① Monopole: Total charge
- ② Dipole: Polarization



[https://en.wikipedia.org/wiki/File:](https://en.wikipedia.org/wiki/File:Ball_and_stick_model_of_a_water_molecule.png)

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Macroscopic Densities

The result (after great suffering) is:

$$\langle \varrho(\mathbf{x}, t) \rangle_M \approx \rho(\mathbf{x}, t) - \nabla \cdot \mathbf{P}(\mathbf{x}, t) + \dots$$

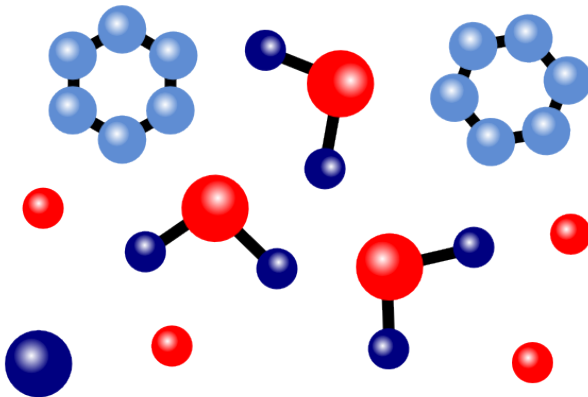
$$\langle \mathbf{j}(\mathbf{x}, t) \rangle_M \approx \mathbf{J}(\mathbf{x}, t) + \frac{\partial \mathbf{P}(\mathbf{x}, t)}{\partial t} + c \nabla \times \mathbf{M}(\mathbf{x}, t) + \dots$$

where

- ρ is the *free charge* density
- $\mathbf{P}$ is the *polarization* density
- $\mathbf{J}$ is the *free current* density
- $\mathbf{M}$ is the *magnetization* density

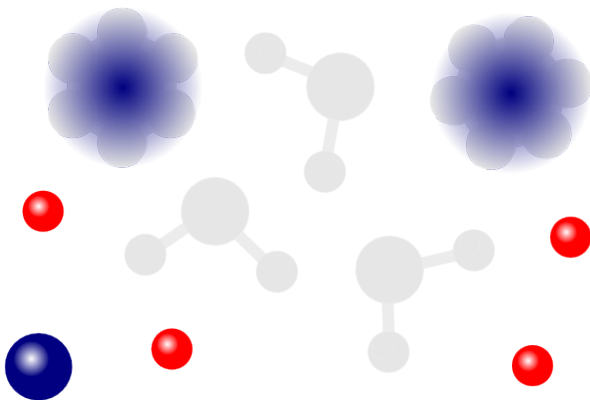
Microscopic Charge Density: ϱ

How much total charge?



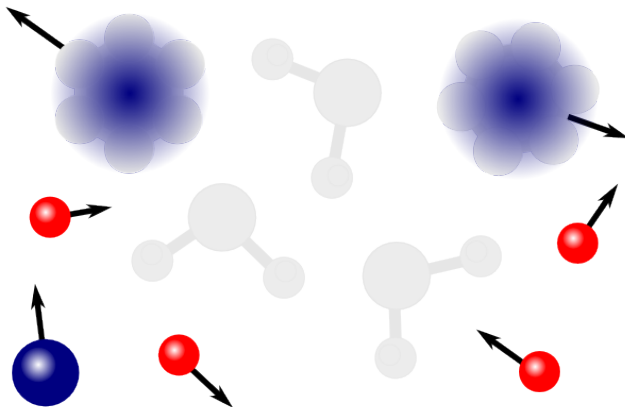
Free Charge Density: ρ

How much mobile charge?



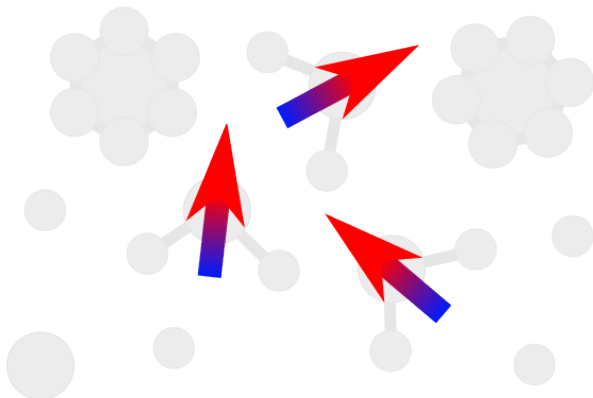
Free Current Density: J

How much mobile charge movement?



Polarization Density: P

How much polarization?

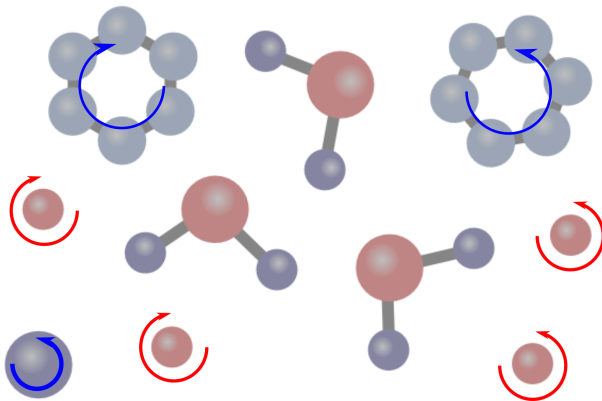


$P(\mathbf{x}, t)$ is the average molecular dipole at $\mathbf{x}$:

$$P(\mathbf{x}, t) \equiv \langle \mu^{\text{mol}} \delta(\mathbf{x} - \mathbf{r}^{\text{mol}}) \rangle_M$$

Magnetization Density: $\mathbf{M}$

How much circulation of charge?



$c\mathbf{M}(\mathbf{x}, t)$ is the average molecular magnetic dipole at $\mathbf{x}$:

$$\mathbf{M}(\mathbf{x}, t) \equiv \frac{1}{c} \left\langle \mu_{\text{mag}}^{\text{mol}} \delta(\mathbf{x} - \mathbf{r}^{\text{mol}}) \right\rangle_M \text{ with } \mu_{\text{mag}}^{\text{mol}} = \sum_n \frac{q_n}{2} \mathbf{r}_n \times \mathbf{v}_n.$$

Read the Signs

	Microscopic	Macroscopic
Electric Field	$\boldsymbol{e}$	$\boldsymbol{E}$
Magnetic Field	$\boldsymbol{b}$	$\boldsymbol{B}$
Charge Density	ϱ	ρ
Current Density	$\boldsymbol{j}$	$\boldsymbol{J}$

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But watch out!

$$E = \langle e \rangle$$

$$\rho \neq \langle \varrho \rangle$$

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$$\begin{aligned} \langle \varrho(\mathbf{x}, t) \rangle_M &\approx \rho(\mathbf{x}, t) - \nabla \cdot \mathbf{P}(\mathbf{x}, t) + \dots \\ \langle \mathbf{j}(\mathbf{x}, t) \rangle_M &\approx \mathbf{J}(\mathbf{x}, t) + \frac{\partial \mathbf{P}(\mathbf{x}, t)}{\partial t} + \dots \end{aligned}$$

Take-Home Points

Free charges can move over large distances

Bound charges are *mostly* stuck in place

The multipole expansion allows us to write $\langle \varrho \rangle_M$ and $\langle \mathbf{j} \rangle_M$ in terms of *macroscopic quantities*:

- The free charge density ρ
- The polarization density $\mathbf{P}$
- The free current density $\mathbf{J}$

Molecular Spectroscopy and Material Polarization

Going Forward: Dielectric Materials

In this course we'll work with *dielectrics*: materials without free charges. *Whole molecules don't move very fast!*

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In dielectrics, both ρ and $\mathbf{J}$ vanish:

$$\nabla \cdot \mathbf{E} = -4\pi \nabla \cdot \mathbf{P}(\mathbf{x}, t)$$

$$\nabla \cdot \mathbf{B} = 0$$

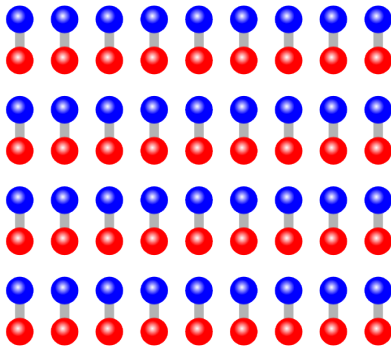
$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0$$

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c} \frac{\partial \mathbf{P}(\mathbf{x}, t)}{\partial t} + 4\pi \nabla \times \mathbf{M}(\mathbf{x}, t)$$

For molecular spectroscopy, the **polarization density** is key.

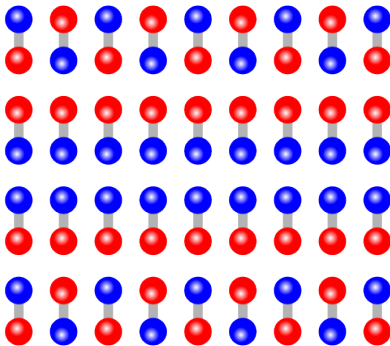
Think about it!

Q: How does polarization contribute to $\langle \varrho \rangle_M$?



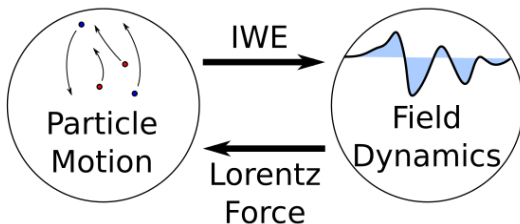
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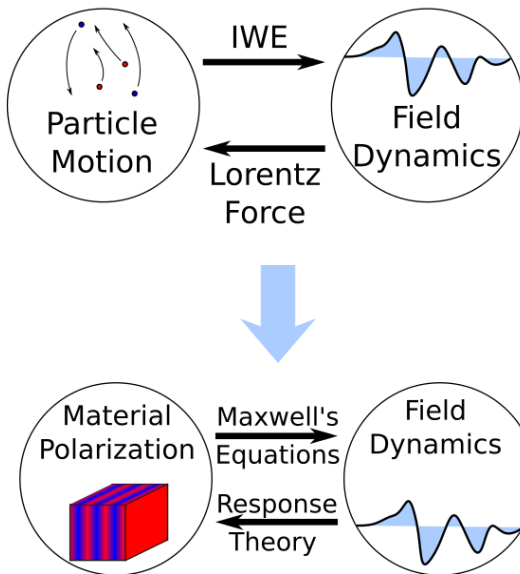


A: It doesn't! It's $\nabla \cdot P$!

Next up: Response Theory



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Take-Home Points

$\mathbf{P}$ doesn't contribute directly to $\langle \varrho \rangle_M$ or $\langle \mathbf{j} \rangle_M$:
It's $\nabla \cdot \mathbf{P}$ and $\frac{\partial \mathbf{P}}{\partial t}$ that contribute.

In macroscopic electrodynamics, *response theory* replaces the Lorentz force law in describing how materials respond to the EM field.